Theory of Computation

BCSE304L — VIT | PYQ Analysis & Exam Practice Guide

Covers: April 2025 · May 2024 · April 2026 Question Papers

Module-wise Weightage (Based on PYQ Pattern)

Module	Topics	Approx Marks	Freq	Option
M1	Intro, Proof Techniques, Grammars, Chomsky Hierarchy	10	High	No
M2	Finite State Automata (DFA, NFA, ε-NFA, Min-DFA)	10	Very High	No
M2	Regular Expr., RE↔Automata, Pumping Lemma	10	Very High	No
M3	Context Free Grammar (CFG, CNF, GNF, CYK, Ambiguity)	10	Very High	No
M3	CFG cont. + Closure Properties / Pumping Lemma CFL	10	High	No
M4	Pushdown Automata (PDA, DPDA, NPDA)	10	High	No
M4	PDA cont. / Mix with CFG	10	Medium	No
M5	Turing Machine (TM types, IO, Acceptor, Encoding, UTM)	10	High	No
M6	Recursive & RE languages, Decidability, PCP	10	High	**Yes (M6/M7)**
M7	Chomsky Hierarchy, Applications	10	Medium	**Yes (M6/M7)**

Note: *Each question carries 10 marks. M6 and M7 have an internal choice (attempt one). All other modules are compulsory. The 2025-26 paper format is taken as the reference.*

Module 1: Introduction to Languages, Grammars & Proof Techniques

CO4

Frequency Analysis

Proof by Contrapositive / Induction	High	*Apr-2025, Apr-2026*
Language Terminologies (Σ, Σ*, strings)	Medium	*Apr-2026*
Types of Grammars & Chomsky Hierarchy	High	*Apr-2025, May-2024, Apr-2026*
Grammar ↔ Language conversion	High	*May-2024, Apr-2026*
String generation from Grammar	Medium	*Apr-2025, Apr-2026*

Important PYQs

Q1.	Let L1 = {n^2 % 3 = 0, n∈Z}, L2 = {n % 3 = 0, n∈Z}. Prove by contrapositive: 'If L1 then L2'.	**[6M]**	Apr-2025
Q2.	Let Σ = {#,S,&}. How many distinct strings of length 5 can be formed over Σ?	**[2M]**	Apr-2025
Q3.	L1 = {w∈Σ* | w begins with a, |w|≥3}. L1-L2 = {aba,abb}. Find L2 using set builder notation.	**[2M]**	Apr-2025
Q4.	Construct CFG for: (i) L = {a^m b^n | n=2m, m≥1} (ii) L = {a^n b^m c^k | k=|n-m|, n,m≥1}.	**[6M]**	May-2024
Q5.	Use induction to show |u^n| = n|u| for all strings u and n.	**[4M]**	May-2024
Q6.	L1={a,ab,ba}, L2={b,aa}. Compute L1L2, L2L1, $L1^3$, L1∪L2, L1∩L2. Which gives max/min strings?	**[10M]**	Apr-2026

Tip: Chomsky Hierarchy diagram (Type 0→1→2→3) is frequently asked in Q8b (Apr-2025) and Q10/12 (May-2024, Apr-2026). Draw it with grammar rules + automaton for each level.

Module 2: Finite State Automata (DFA / NFA / ε-NFA / Minimization) CO1

Frequency Analysis

Language → DFA construction	Very High	*Every paper*
NFA construction from Language	Very High	*Every paper*
ε-NFA → NFA (epsilon elimination)	High	*May-2024 B1+TB1*
NFA → DFA (subset construction)	High	*May-2024 B2+TB2, Apr-2026*
DFA Minimization (Table-filling)	Very High	*May-2024 (both slots), Apr-2026*
Reversal / Complement of DFA/NFA	High	*May-2024, Apr-2025*
DFA cross-product (∩, ∪)	Medium	*May-2024 B1+TB1*
Minimized DFA states for length patterns	Medium	*Apr-2025*

Important PYQs

Q1.	Construct NFA over Σ={a,b} accepting all strings whose length is divisible by 5. Then convert to equivalent DFA.	**[5+5M]**	May-2024 B2
Q2.	Minimize the given 7-state DFA using Table Filling Method (states 1-7 with transitions on a,b).	**[10M]**	May-2024 B1
Q3.	Convert ε-NFA (q0,q1,q2 with ε,a,b,c transitions) to NFA without ε-moves.	**[10M]**	May-2024 B1
Q4.	NFA P with 7 states (p0-p6), Σ={0,1,ε}. Find equivalent DFA. Is DFA more powerful than NFA?	**[10M]**	Apr-2026
Q5.	Construct DFA for L={w | w ends with 101} over Σ={0,1}. Give transition table.	**[5M]**	Apr-2026
Q6.	Construct NFA for L={w | w contains 'abba' or 'bab'} over Σ={a,b}.	**[5M]**	Apr-2026
Q7.	Statement analysis: (1) Reverse of DFA → DFA or NFA? (2) Complement of DFA → complemented language? (3) Complement of NFA → always complemented?	**[5M]**	Apr-2025
Q8.	How many states for minimized DFA accepting: exactly n-length / at-least n-length / at-most n-length strings?	**[5M]**	Apr-2025

Q9.	L1={w | |w| is odd}, L2={w | |w| divisible by 3}. Find M1×M2, complement of M1×M2, reversal of M1×M2.	**[10M]**	May-2024 B1

Module 3: Regular Expressions & Regular Languages — CO2

Frequency Analysis

RE → NFA (Thompson construction)	Very High	*Every paper*
DFA → RE (Arden's / State Elimination)	High	*May-2024 B1*
Regular Grammar construction	High	*May-2024 B2*
Pumping Lemma (irregularity proof)	Very High	*May-2024 B1+TB1, Apr-2026*
Closure Properties of Regular Languages	Medium	*Apr-2025*
RE for language without consecutive a's etc.	High	*Apr-2025*

Important PYQs

Q1.	Write RE for: 'no two consecutive a's' over Σ={a,b}. Then convert it to NFA.	**[3+7M]**	Apr-2025
Q2.	R = (aa)*+ ba■+(ab)*. (a) Construct NFA. (b) Construct regular grammar G for aa*(ab/a)*.	**[5+5M]**	May-2024 B2
Q3.	Derive RE from given DFA using Arden's Lemma AND State Elimination method.	**[10M]**	May-2024 B1
Q4.	Convert (a+b)*aba into NFA.	**[5M]**	Apr-2026
Q5.	Prove L = {a^(2n) b^n a^n | n≥0} is not regular using Pumping Lemma.	**[5M]**	Apr-2026
Q6.	Prove L = {a^n b^l a^k | n=l or l=k} is not regular using Pumping Lemma.	**[5M]**	May-2024 B1
Q7.	Construct DFA for L={w | w does not contain 'aa' or 'bb'}. Derive RE from it.	**[10M]**	May-2024 B2
Q8.	FSA accepting L={n0(w) mod 2=0 AND n1(w) mod 2=0} → convert to regular grammar (start A).	**[5M]**	May-2024 B1

Module 4: Context Free Grammars (CFG, CNF, GNF, CYK, Ambiguity) CO3

Frequency Analysis

CFG → CNF (Chomsky Normal Form)	Very High	*Every paper*
CFG → GNF (Greibach Normal Form)	Very High	*Apr-2025, Apr-2026*
CYK Membership Algorithm	Very High	*May-2024 B2, May-2024 B1, Apr-2026*
Ambiguity in CFG	High	*May-2024 B2, Apr-2026*
Simplification (null, unit, useless productions)	High	*May-2024 B1, Apr-2026*
Pumping Lemma for CFL	Medium	*Apr-2026*
CFG for complex languages (a^n b^m c^k type)	High	*May-2024 B2, Apr-2026*
LMD / RMD / Parse Trees	Medium	*Apr-2025, Apr-2026*

Important PYQs

Q1.	CFG: S→SS|aSb|ab. Convert to GNF. Prove 'abab' derivable in 4 steps via RMD using GNF.	**[10M]**	Apr-2025
Q2.	CFG: G={(S,A,B),(a,b), S→ASB|ε, A→aAS|a, B→SbS|A|bb}. Convert to CNF.	**[6M]**	May-2024 B2
Q3.	Check ambiguity: K→QK|ε, Q→Qa|aQb|ab. Show derivation steps (K is start).	**[4M]**	May-2024 B2
Q4.	CYK: w='He eats a fish with a fork'. Grammar: S→NP VP, VP→VP PP|V NP|eats, etc.	**[10M]**	May-2024 B2
Q5.	CYK for w=babab. S→AC|CB, A→CA|a, B→AB|b, C→CB|b.	**[10M]**	May-2024 B1
Q6.	CFG G: A→BAZ, B→aBA|a|ε, Z→AbA|B|bb. Convert to CNF (start symbol A).	**[5M]**	May-2024 B1
Q7.	Prove G has ambiguity for w=aaaabbbb. S→ASB|BSA|AB|BA, A→Aa|a, B→Bb|b.	**[5M]**	May-2024 B1
Q8.	CFG: S→ABA|B|ε, A→00|ε, B→11. Find nullable/unit productions. Convert to CNF then GNF.	**[10M]**	Apr-2026

Q9.	Construct CFG for L={a^i b^j c^k | i,j,k≥0, i=j or j=k}. Give leftmost derivation + parse tree.	**[10M]**	Apr-2026

Module 5: Pushdown Automata (PDA / DPDA / NPDA) CO1

Frequency Analysis

Language → PDA (by empty stack)	Very High	*Every paper*
Language → PDA (by final state)	High	*Apr-2025, May-2024 B1*
ID (Instantaneous Description) trace	Very High	*Apr-2025, May-2024 B1+B2*
DPDA vs NPDA justification	High	*Apr-2025*
CFL → PDA (from grammar)	Medium	*May-2024 B1*

Important PYQs

Q1.	Design PDA for L={a^i b^j c^k d^l | i==k or j==l, i,j,k,l>0}. Is it DPDA or NPDA? Justify.	**[10M]**	Apr-2025
Q2.	Construct PDA for L={a^n b^(m+n) c^m | n≥0, m≥1} by empty stack. Give ID for 'aaabbbbbcc'.	**[10M]**	May-2024 B2
Q3.	Construct PDA for L={a^n b^m c^n | n,m≥1} by final state. Show moves for 'aabbbbcc'.	**[10M]**	May-2024 B1
Q4.	Construct PDA for L={a^(2m) b^n c^(n) d^(2m) | m,n≥0}. Give ID for one accepted string.	**[10M]**	Apr-2026

Module 6: Turing Machines (TM Types, IO, Acceptor, UTM, Halting) CO1

■ M6 and M7 carry an internal OPTION — you must attempt ONLY ONE of the two.

Frequency Analysis

TM as IO device (Copy-Paste, transducer)	Very High	*Apr-2025, May-2024 B2*
TM as Acceptor (language recognition)	High	*May-2024 B1, Apr-2026*
TM computation function (monus, etc.)	High	*Apr-2025 9a*
Encoding of TM / Universal TM	High	*May-2024 B2 Q10, May-2024 B1 Q10*
Halting Problem (undecidability)	High	*Apr-2026 9b*
Types of TM (multi-tape, non-deterministic, etc.)	Medium	*Apr-2026 9a-ii*
Unary to Ternary converter TM	Medium	*Apr-2025 9b*

Important PYQs

Q1.	Design TM for Copy-Paste: input w→output ww over Σ={0,1}. Explain transitions for example 3 or 4 (input 00→0000, input 111→111111).	**[10M]**	Apr-2025
Q2.	Design TM: f(m,n) = m-n if m>n, else blank. Compute ID for f(0000,00) and f(00,000). [Unary subtraction]	**[10M]**	Apr-2025 9a
Q3.	Design TM for unary-to-ternary converter. Examples: 00000→12, 000→10, 000000000000000→120.	**[10M]**	Apr-2025 9b
Q4.	Obtain the CODE for Universal TM ■M,1010■. M has transitions: δ(q1,1)=(q3,0,R), δ(q3,0)=(q1,1,R), etc.	**[6M]**	May-2024 B2
Q5.	Construct TM for L={a^n b^n c^n | n≥1}. Show ID from abc to $a^2b^2c^2$.	**[10M]**	May-2024 B1
Q6.	Design TM accepting L={a^n b^(2n) | n≥1}.	**[5M]**	Apr-2026 9a-i
Q7.	Analyze the TM Halting Problem. Discuss why it cannot be solved by any TM.	**[10M]**	Apr-2026 9b
Q8.	TM transducer: input w∈{a,b}* → output ww^R. Example: w=aba → output abaaba.	**[10M]**	May-2024 B2 Q9

Q 9.	Construct TM for L={a^n b^m a^(n+m) | n≥0, m≥1}. Show ID.	**[10M]**	May-2024 B1 Q11

Module 7: Recursive/RE Languages, Decidability, PCP, Chomsky Hierarchy CO2

■ **M6 and M7 carry an internal OPTION — you must attempt ONLY ONE of the two.**

Frequency Analysis

Recursive & RE languages (properties, proofs)	Very High	*Apr-2025 Q6, May-2024 B1 Q12*
Post Correspondence Problem (PCP)	High	*Apr-2025 Q8a, May-2024 B2 Q10b*
Chomsky Hierarchy (draw + interpret)	Very High	*Apr-2025 Q8b, May-2024 B1 Q12, Apr-2026*
Decidability vs Undecidability	High	*May-2024 B1 Q12, Apr-2026*
Closure properties of recursive / RE languages	High	*May-2024 B1 Q11*
L(G1) ∩ L(G2) = ∅ undecidability proof	Medium	*May-2024 B1 Q12b*

Important PYQs

Q1.	Prove/Disprove: (i) If L and L■ are RE → both recursive? (ii) If L is recursive → L■ is recursive? (iii) If L and L■ are recursive → both RE?	**[3+4+3 M]**	Apr-2025 Q6
Q2.	PCP: List A={1,10111,10}, List B={111,10,0}. Does it have a solution? Justify.	**[6M]**	Apr-2025 Q8a
Q3.	Draw Chomsky Hierarchy. Interpret relationship among grammars, languages, and automata.	**[4M]**	Apr-2025 Q8b
Q4.	PCP: x={b,bab3,ba}, y={b3,ba,a}. Does it have a solution?	**[4M]**	May-2024 B2 Q10b
Q5.	Discuss recursive and recursively enumerable languages with definitions and examples.	**[4M]**	May-2024 B1 Q12a
Q6.	G1 is unrestricted grammar, G2 is regular grammar. Show L(G1)∩L(G2)=∅ is undecidable.	**[6M]**	May-2024 B1 Q12b
Q7.	Discuss any two closure properties where recursive languages are CLOSED and two where RE languages are NOT CLOSED.	**[10M]**	May-2024 B1 Q11 (i+ii)
Q8.	Construct and explain Chomsky Hierarchy: language types, automata, grammar rules, effect of computational power.	**[10M]**	Apr-2026 Q10a

Q9.	Critically analyze Post Correspondence Problem and how it leads to undecidability in formal language theory.	**[10M]**	Apr-2026 Q10b

Quick Reference: Key Formulas & Theorems

Strings & Languages

- Σ^* = all strings including ε over Σ
- $|\Sigma| = k \to k^n$ strings of length n
- $L1 \cdot L2 = \{xy \mid x \in L1, y \in L2\}$ (concatenation)
- $L^* = \{\varepsilon\} \cup L \cup L^2 \cup \ldots$ (Kleene closure)

DFA Minimization (Table-Filling)

- Step 1: Mark all (accept, non-accept) pairs as distinguishable
- Step 2: For each unmarked (p,q), if $\delta(p,a)$ and $\delta(q,a)$ are distinguishable $\to$ mark (p,q)
- Step 3: Repeat until no change. Merge all unmarked pairs.
- Dead state: unreachable / trap state — include in product but can remove from minimized DFA

Pumping Lemma — Regular Languages

- For regular L, $\exists$ pumping length p such that
- Any $w \in L$ with $|w| \geq p$ can be split: $w=xyz$, $|xy| \leq p$, $|y| \geq 1$
- And $\forall i \geq 0$: $xy^i z \in L$
- To prove NON-regular: choose w, show no split satisfies the condition

CNF (Chomsky Normal Form)

- Every production: $A \to BC$ or $A \to a$
- Steps: (1) Remove ε-productions (2) Remove unit productions
- (3) Remove useless productions (4) Convert long productions

GNF (Greibach Normal Form)

- Every production: $A \to a\alpha$ where $a \in \Sigma$, $\alpha \in V^*$
- Steps: (1) Order variables (2) Eliminate left recursion
- (3) Substitute to ensure first symbol is terminal

CYK Algorithm

- Grammar must be in CNF first
- Build table T[i][j] = set of variables that derive w[i..i+j-1]
- Base: $T[i][1] = \{A \mid A \to w_i\}$
- Recur: $T[i][j] = \{A \mid A \to BC, B \in T[i][k], C \in T[i+k][j-k]$ for some $k\}$
- $w \in L(G)$ iff $S \in T[1][n]$

Pumping Lemma — CFLs

- For CFL L, $\exists$ p such that any $w \in L$, $|w| \geq p$ can be split: $w=uvxyz$
- $|vy| \geq 1$, $|vxy| \leq p$
- $\forall i \geq 0$: $uv^i xy^i z \in L$
- To prove NON-CFL: choose w cleverly (e.g., $a^p b^p c^p$)

PDA Key Points

- PDA = $(Q, \Sigma, \Gamma, \delta, q0, Z0, F)$
- $\delta: Q \times (\Sigma \cup \{\varepsilon\}) \times \Gamma \to 2^{(Q \times \Gamma^*)}$
- Accept by empty stack: $F=\varnothing$, stack empties on acceptance

- Accept by final state: reach a state in F
- DPDA $\subset$ NPDA (deterministic is strictly less powerful for CFLs)

Turing Machine Key Points

- TM = $(Q, \Sigma, \Gamma, \delta, q0, B, F)$ — B is blank symbol
- δ: $Q \times \Gamma \rightarrow Q \times \Gamma \times \{L,R\}$
- Configuration (ID): $\alpha q \beta$ — q is current state, tape = $\alpha\beta$
- TM halts in accept state $\rightarrow$ string accepted
- Universal TM: reads ■M,w■ and simulates M on w

Decidability Quick Reference

- Recursive (Decidable): TM halts on ALL inputs (accept or reject)
- RE (Semidecidable): TM halts on accepted inputs, may loop on others
- Co-RE: complement is RE
- Halting Problem: undecidable — no TM can decide if arbitrary TM halts
- PCP: undecidable — existence of matching sequence in two lists
- If L is recursive $\rightarrow$ L■ is recursive (closed under complement)
- If L and L■ are both RE $\rightarrow$ L is recursive

Exam Strategy & Last-Minute Tips

Always-Attempted Topics (Every Paper)	DFA minimization, NFA construction, RE↔automata, CFG→CNF, PDA design with ID trace.
High-Value Topics	GNF conversion, CYK algorithm, TM as IO device. Each is ~10 marks and formulaic.
Option Strategy (M6/M7)	If you're strong on TM design → pick M6. If you're better at decidability theory + Chomsky hierarchy → pick M7. Practice BOTH enough to make the call in exam.
ID Traces	Always write ID step-by-step. Even if your PDA/TM design has a small error, you get partial marks for correct ID format.
Proofs	Pumping Lemma proofs: always choose a specific pumping string that breaks all cases. For contrapositive proofs: negate the conclusion, derive contradiction.
CNF/GNF Pitfall	In CNF, ε-productions are removed FIRST. In GNF, handle left recursion carefully — introduce new variables for right-recursive version.
CYK Tip	Draw the full triangle table. Label rows and columns clearly. State the conclusion: 'S ∈ T[1][n]? YES/NO → w ∈/∉ L(G)'.
PCP Tip	Try to find a solution by matching prefixes. If you need to prove it has NO solution, argue why no finite concatenation can match.
Time Management	Spend ~15 min per 10-mark question. Leave 5 min to review IDs and derivations. Diagram questions (DFA, PDA, TM) should be drawn clearly with all states labeled.

All the best for your exam! ■ — BCSE304L TOC PYQ Guide